RoadFury: Transformer-Based Test Prioritization for Simulation-Based Testing of Self-Driving Cars

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***Abstract*—We present RoadFury, a test prioritization tool for simulation-based regression testing of self-driving cars. In BeamNG.tech simulations, executing large test suites is prohibitively expensive; prioritization reorders tests so that failure-revealing cases execute first. RoadFury extracts a 10-channel sequential feature matrix from road geometry—curvature, heading sin/cos, curvature jerk, curvature acceleration, local variability, and positional encodings—and processes it with a Pre-LN Transformer Encoder with learnable `[CLS]` token, trained with Stochastic Weight Averaging (SWA) for robust generalization. Deployed as a Docker-containerized gRPC service for the ICST 2026 SDC Testing Competition [7], RoadFury achieves APFD = 0.8042 ± 0.0120, surpassing ITEP4SDC (0.7812) and random ordering (0.4930).**

***Index Terms*—test prioritization, self-driving cars, Transformer, stochastic weight averaging, road geometry**

I. Introduction

Simulation-based testing of SDCs in high-fidelity simulators (e.g., BeamNG.tech) is indispensable for validation yet expensive [1], [2]: individual tests run for tens of seconds to minutes. The rapid ascent of AI agents in autonomous navigation [11] widens this gap further. *Test prioritization* addresses this by finding a permutation π that maximizes early fault detection, measured by APFD [3]:

$$\text{APFD}(\pi) = 1 - \frac{\sum_{i=1}^{m} TF_i}{n \cdot m} + \frac{1}{2n} \tag{1}$$

where n is total tests, m failing tests, TF_i the rank of the i-th fault (1.0: all failures first; 0.5: random).

Prior tools—SDC-Scissor [2] and ITEP4SDC—use 3 hand-crafted features (segment length, angle change, curvature) in feedforward networks, discarding the *sequential ordering* of road geometry. **RoadFury** addresses this gap with: (1) a *10-channel geometric feature representation* encoding higher-order curvature dynamics and positional context; and (2) a *Transformer Encoder with SWA* capturing long-range inter-point dependencies. A systematic evaluation [10] confirms Transformers consistently outperform classical classifiers on sequential feature inputs; analogous early-stage ML prediction of AI-artifact outcomes is validated in related work [9]. SDC-Scissor [1], [2] pioneered road-feature ML; Yoo and Harman [3] survey regression testing; Izmailov et al. [5] show SWA finds wider optima. To identify the optimal paradigm, we systematically compare visual CNNs, GNNs, zero-shot LLMs, and sequential Transformer encoding—empirically converging on the latter.

TABLE I
10-Channel Road Geometry Features

Ch.	Feature	Description
f_0	Segment length	$\lVert p_{i+1} - p_i \rVert_2$
f_1	Abs. angle change	$\lvert \Delta\theta_i \rvert$
f_2	Menger curvature	$\kappa_i = 4A_i/(a_i b_i c_i)$, Heron's formula
f_3	Curvature jerk	$\Delta\kappa_i$, 1st derivative of κ
f_4	Cum. dist. norm	$\sum_{j<i} \lVert p_{j+1} - p_j \rVert / d_{\text{tot}} \in [0,1]$
f_5	Heading sin	$\sin(\theta_i)$, $\theta_i = \text{atan2}(\Delta y, \Delta x)$
f_6	Heading cos	$\cos(\theta_i)$, avoids $\pm\pi$ wrap
f_7	Rel. position	$i/(L-1) \in [0,1]$
f_8	Local curv. std	$\text{std}(\kappa_{i-5:i+5})$, window = 11
f_9	Curv. accel.	$\Delta^2\kappa_i$, 2nd derivative of κ

II. Approach

A. Road Normalization and Feature Extraction

Each test case is a variable-length sequence of 2D road control points. We resample all sequences to L=197 points via linear interpolation and extract 10 intrinsic geometric channels (Table I), yielding a (197×10) translation- and rotation-invariant feature matrix, z-normalized with training-set statistics.

B. RoadTransformer Architecture

The (197×10) matrix passes through four stages (Fig. 1). **(1+2) Project + embed:** $\mathbf{h}_i = \text{GELU}(\text{LN}(\mathbf{W}_p\mathbf{f}_i + \mathbf{b}_p))$, projecting 10→128; learnable `[CLS]` $\in \mathbb{R}^{128}$ prepended; positional embeddings added (len. 198).

(3) Pre-LN Transformer Encoder [8]: 4 layers, 8 heads, FFN 512, GELU, dropout 0.1. Pre-LN residual updates: $\mathbf{Z}' = \mathbf{Z}^{(l)} + \text{MHA}(\text{LN}(\mathbf{Z}^{(l)}))$; $\mathbf{Z}^{(l+1)} = \mathbf{Z}' + \text{FFN}(\text{LN}(\mathbf{Z}'))$. CLS output → LN → Linear(128→64) → GELU → Drop(0.2) → Linear(64→1) → Sigmoid. Output $\hat{y} \in [0,1]$ is failure probability (∼400K params total).

C. Training (SWA) and Deployment

Adam (lr = 1e-3, BCE), 75 epochs on 28,804 samples. SWA [5] starts at epoch 50—averaging weights over the final 25 epochs finds a flatter loss basin, improving robustness to unseen road geometries. The SWA checkpoint (`roadfury_best.pt`) stores the averaged state dict and per-channel normalization statistics.

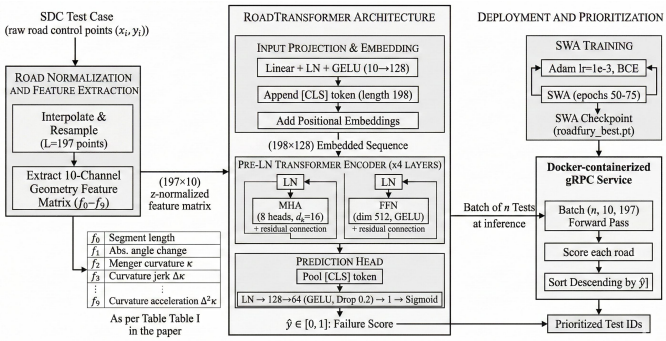

Fig. 1. RoadFury Pipeline: Road Polyline → Prioritized IDs

Fig. 1. RoadFury pipeline: road polyline → (197×10) feature matrix → Pre-LN Transformer Encoder with `[CLS]` → MLP head → failure score $\hat{y}$ for sorting.

TABLE II
MULTI-PARADIGM COMPARISON (956 COMP. TESTS, 30 TRIALS)

Method	APFD	Paradigm
Random	0.493	–
LLM zero-shot	0.487 ± 0.019	Qwen2.5-7B, 4-bit
GNN (road graph)	0.533 ± 0.025	3-layer GCN+KNN
ResNet50→XGB	0.572 ± 0.013	Road image CNN
CNN+Handcraft (best)	0.673 ± 0.017	ResNet50+44-feat
ExtendedCNN-10ch	0.768 ± 0.014	10-ch CNN, 28K data
ITEP4SDC	0.781	ONNX MLP, 3-feat
RoadFury (ours)	$\mathbf{0.804 \pm 0.012}$	**Transformer+SWA**

At inference, n tests are batched $(n, 10, 197)$, scored in one forward pass, and sorted by descending $\hat{y}$. The service exposes `Name()`, `Initialize(oracle)`, and `Prioritize(tests)`→scores via gRPC, containerized on port 4545.

III. EVALUATION

Dataset. SensoDat [6]: 36,006 labeled BeamNG tests, split 80/20 (28,804 train / 7,202 val); 956-test competition set is the held-out target.

Baselines. Beyond Random and ITEP4SDC (prior SoA, ONNX MLP, 3 features), we evaluate three additional paradigms on the same 30-trial protocol (Table II): (i) **LLM zero-shot**—Qwen2.5-7B [11] in 4-bit quantization scores each road from a textual curvature summary; (ii) **GNN**—road points as a graph with sequence and spatial KNN edges (3-layer GCN [16]); (iii) **road image CNN**—road polylines rendered to 224×224 px images, features extracted by ImageNet-pretrained CNNs [4] fed into XGBoost.

Table II reveals a clear *paradigm hierarchy*. Zero-shot LLMs score below random (0.487 ± 0.019): language models cannot reliably reason about curvature thresholds. GNNs reach 0.533 ± 0.025—spatial connectivity misses the sequential ordering critical for vehicle dynamics. Visual CNNs plateau at 0.673 even fused with 44 handcrafted features. The *ExtendedCNN-10ch*—our 10-channel featuriser on a deeper CNN trained with 28K SensoDat data—reaches 0.768 ± 0.014, showing *feature quality* matters more than scale. **RoadFury** achieves $\mathbf{0.804 \pm 0.012}$: +2.9% over ITEP4SDC, +4.6% over ExtendedCNN-10ch, +63% over random.

Ablation (single-factor). Isolating each modification over the base Pre-LN Transformer (0.790): Multi-Scale Conv Stem: 0.767; DropPath [15]: 0.768; Triple Pool: 0.768; Mixup [14]: 0.773; TTA: 0.770; Focal Loss [13] (γ=2.0): 0.782 ± 0.014; **SWA** [5]: $\mathbf{0.804 \pm 0.012}$. SWA is the sole consistently beneficial modification (+0.014); all structural variants hurt the base Pre-LN design, which is already well-matched to sequential road geometry.

IV. CONCLUSION

RoadFury achieves $\text{APFD} = 0.804 \pm 0.012$ on the ICST 2026 SDC Competition benchmark—surpassing all four explored paradigms: zero-shot LLMs (0.487, *below* random), GNNs (0.533), visual CNNs (0.673), and the prior SoA ITEP4SDC (0.781). Our systematic exploration empirically validates that sequential geometric encoding—not visual, graph, or language representations—is the dominant inductive bias for SDC test prioritization, and that SWA over a Pre-LN Transformer is the most effective training strategy. Future work targets online oracle adaptation and multi-scale SWA scheduling. Tool: https://github.com/chisngyen/sdc-test-prioritization-novel.

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